Open problems:
Directional vectors vs quantum state vectors

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`). Items marked [BLOCKS VERIFICATION] stand in the way of checking a main claim of the brief as the brief states it: a *formalization gap* can be closed in Lean alone, while a *brief issue* needs the brief to be corrected or made precise first.

No item below blocks the verification of a main claim of the brief.

From the brief

1. **Why the probability is affine.** The brief requires that "for any other state, the probability of outcome must be proportional to the component in that direction" and reads off eq. (2) from Fig. 2. The requirement is stated, not derived. A precise version, that an affine function of the direction which is invariant under rotations about z and takes the values 1 at e_z and 0 at $-e_z$ is $(1 + v^z)/2$, is not formalized; only the resulting law is (`born_stateVector`). (Brief, Sec. 2, before eq. (2).)

2. **Half a turn and rotations.** "A line returns to itself after half a turn" is formalized only as $\theta \mapsto \theta + 2\pi$ (`stateVector_add_two_pi`). The statement behind it, that a rotation of the direction by an angle α acts on state vectors by a unitary of half the angle, and that $SU(2) \to SO(3)$ is two-to-one, is not formalized. (Brief, Sec. 2, last paragraph.)

3. **Beyond the qubit.** The introduction says in general that quantum states are rays, with linearity in terms of probability. Only the two-state system is formalized, where rays are directions in space; for more levels the pure states are not a sphere of directions, and the brief does not say what replaces eq. (1). (Brief, Sec. 1.)

From the formalization

1. **The projective line as the sphere.** That rays of $\mathbb{C}^2$ correspond one to one to unit vectors of $\mathbb{R}^3$ is proved pointwise (`exists_eq_smul_iff`, `exists_eq_smul_stateVector`, `exists_direction_eq`). It is not stated as an equivalence, or a homeomorphism, between Mathlib's `Projectivization` of $\mathbb{C}^2$ and the unit sphere. (Log, "Differences".)